

Flatland Exploration Vehicle

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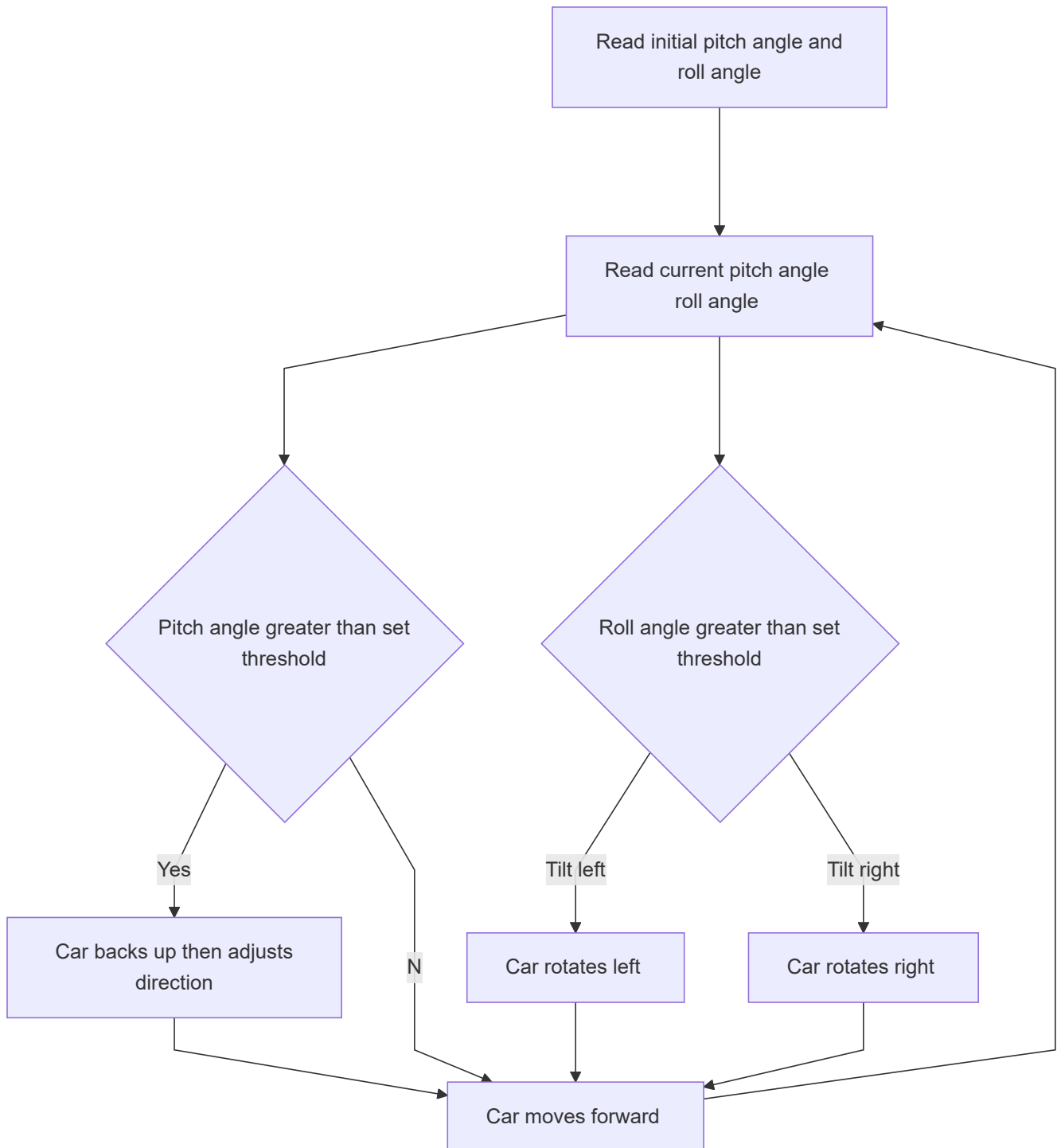
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Function Description

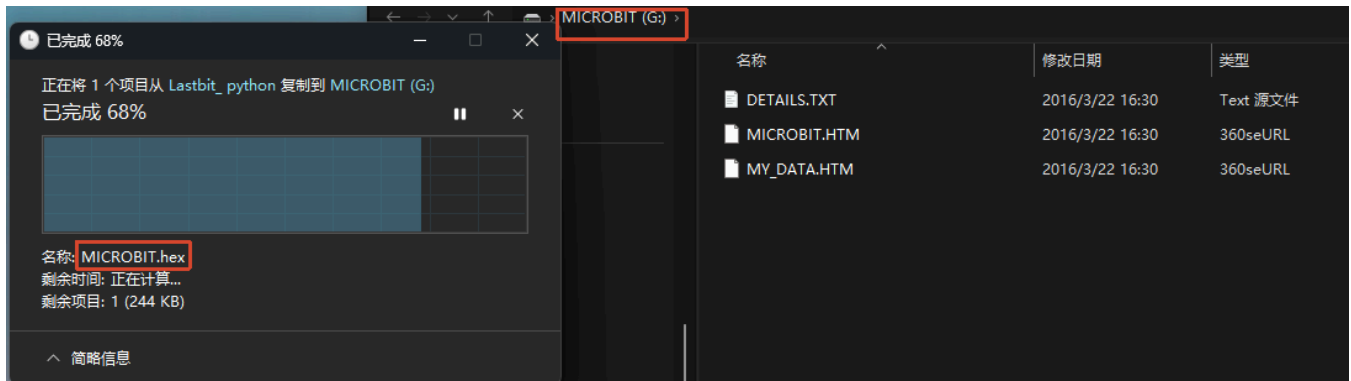
In this section, we will learn to use the accelerometer sensor to detect tilt angles and implement the function of automatically finding flat ground. The car will determine whether it is currently level based on the pitch and roll angles. If it is tilted, it will automatically adjust its direction to find flat ground.

Working Principle



Experiment Steps

Make sure `LASTBIT.hex` is flashed before flashing, otherwise an error will occur and the module cannot be imported



1. Before flashing, switch the switch to the OFF position and use a microUSB data cable to connect the computer and the micro:bit main board. **Note that it is the Microbit interface, not the Charging port on the expansion board.**
2. Open the Mu software. Here you can directly copy the program source code we provided; the operation steps are as follows. You can also enter the code manually in the editing window. Note! All English letters and symbols should be entered in English input mode. Use the Tab key for indentation, and the last line should end with a blank program.

Click the **New** button in the toolbar at the top left, and a file titled "untitled" will appear.



3. Copy the content of the **Code Implementation** section provided below into the current file. You will notice a red dot after the title bar, indicating that the code content has been modified but is in an unsaved state.

Whether to save or not does not affect code checking or flashing.

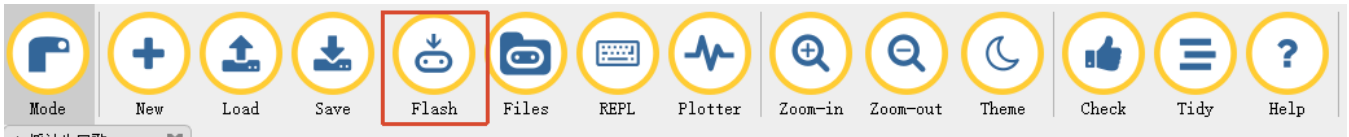
4. Now you can click **check** to verify whether there are any problems with the code format.



5. Now if you have already connected the MICROBIT to the PC in advance following the steps above, and flashed **LASTBIT.hex**, the next step is to download the current program to the Microbit.

Click the **Flash** tool in the toolbar to start flashing!

During flashing, the orange-yellow indicator light near the Microbit main board will blink frequently. When the editor shows **Copied code onto micro:bit.** at the bottom, it means the flashing is complete, and the indicator light will no longer blink after flashing.



6. Unplug the microUSB data cable and switch the switch to the ON position.

Code Implementation

```
# 12. Flat Ground Explorer
from microbit import *
from lastbit import LASTBIT

Lastbit = LASTBIT()

# Initialize servos
Lastbit.set_all_servos(90)
sleep(1000)
Lastbit.set_servo('S2', 150)
sleep(1000)

while True:
    # Get pitch and roll angles
    pitch = Lastbit.get_pitch()
    roll = Lastbit.get_roll()

    # print("Pitch:", pitch, "Roll:", roll)

    # Tilt up
    if pitch > 15:
        display.show(Image.ARROW_N)
        Lastbit.car_control('BACKWARD', 100, 500)
        Lastbit.car_control('SPINLEFT', 150, 300)
    # Tilt left
    elif roll > 15:
        display.show(Image.ARROW_E)
        Lastbit.car_control('SPINLEFT', 150, 800)
    # Tilt right
    elif roll < -15:
        display.show(Image.ARROW_W)
        Lastbit.car_control('SPINRIGHT', 150, 800)
    else:
        display.show(Image.ARROW_S)

    Lastbit.car_control('FORWARD', 130, 500)
```

Experimental Phenomenon

The car will detect its current angle information. Lift the front of the car, and when the detected lift angle reaches the threshold, the car will back up, turn left to adjust its heading, and then move forward to find flat ground. When tilted to the left, the car will turn left to adjust its heading; when tilted to the right, the car will turn right to adjust its heading, then move forward, just like an explorer looking for flat ground. The angle threshold can be adjusted according to the actual situation.

Notes

- The tilt detection threshold can be adjusted according to the actual situation
- This case depends on the terrain, and the threshold requires multiple debugging sessions. It is recommended to use a thin book to simulate a slope road condition for the experience